ARDA BASARICI

AI Engineer — LLM systems, evaluation-first

Software engineer, 3 years of professional experience · Mathematics · Turkey (UTC+3) · open to remote and relocation
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SUMMARY

Software engineer with 3 years of professional experience shipping production games (promoted to Senior at Tiplay Studio) and a mathematics background. In 2026 built and deployed two LLM applications solo, an agent (leave-agent.ardabasarici.dev) and a review-analysis app (steamlens.ardabasarici.dev), owning each end to end: evaluation data, backend, spend controls and infrastructure.

TECHNICAL SKILLS

AI / LLM	AI agents and agentic workflows (LangGraph, tool calling, human-in-the-loop approval), LLM evaluation (gold sets, benchmarks, LLM-as-judge), prompt engineering, structured outputs, guardrails, AI-assisted development (Claude Code)
Languages	Python, TypeScript / JavaScript, C#, SQL
Backend / data	FastAPI, PostgreSQL, SQLite, REST APIs, WebSockets, pandas, NumPy, pytest
Infrastructure	AWS (EC2, IAM, SSM, CloudWatch, Bedrock), Docker, Terraform, Ansible, GitHub Actions CI/CD, monitoring and alerting, Cloudflare, Linux
ML / mathematics	PyTorch, scikit-learn, reinforcement learning, probability and statistics, experiment design
Spoken languages	Turkish (native), English (full professional)

EXPERIENCE

Tiplay Studio — Senior Game Developer Feb 2022 – Dec 2024

Remote · joined as Game Developer, promoted to Senior · ~5-person web-games team

- **Delivery:** Shipped 20+ production games in TypeScript / JavaScript and C# (Cocos Creator, PlayCanvas, Unity WebGL) for web, in-car and TV platforms (AirConsole, including BMW / VW in-car; CrazyGames; Famobi; Snapchat); reviewed and pair-programmed across the team's projects and developed platform approaches and workarounds reused by other developers.
- **Real-time backend (WebSocket):** Designed and shipped AirConsole's first car-to-car multiplayer system, including the server and WebSocket backend the platform did not provide, from experiment to production.
- **Product ownership (TypeScript, Cocos Creator):** Developed Card Lounge largely solo: a live multiplayer card-game hub for AirConsole's in-car platform, with bots for single-player (the poker bot estimates hand strength by Monte Carlo simulation); over a year in production.
- **Partner-facing work:** Took part in most partner meetings (AirConsole, TikTok, Famobi, CrazyGames, Snapchat), held in English. AirConsole sent two other studios' projects to Tiplay to be fixed, mostly integration problems. At Gamescom 2024, on the stand, fixed a live production issue in a Tiplay game on an untested BMW model.

Career break: independent Unity project, compulsory military service, move to AI engineering Jan 2025 – May 2026

Boğaziçi University — Teaching Assistant, Department of Mathematics 2019 – 2022

- Led weekly problem sessions and office hours in 7 undergraduate courses over 5 semesters; graded exams.

EDUCATION

Boğaziçi University — B.Sc. Mathematics (English-medium) 2014 – 2018

GPA 3.40 / 4.00, Honors Certificate; third-highest GPA in the Mathematics cohort · TÜBİTAK undergraduate fellowship

Boğaziçi University — Graduate coursework in Mathematics (M.Sc. program) 2018 – 2022

GPA 3.92 / 4.00; coursework complete, thesis not submitted · TÜBİTAK graduate fellowship 2018 – 2020

INDEPENDENT PROJECTS

Leave Impact Agent — LLM agent that investigates a leave and drafts a coverage plan Aug – Nov 2026

Python · LangGraph · Amazon Bedrock (Claude Haiku 4.5) · PostgreSQL · FastAPI · Docker · GitHub Actions · AWS
leave-agent.ardabasarici.dev · github.com/arda-basarici/leave-impact-agent · evaluation report (PDF)

- **LLM agent (LangGraph, tool calling):** Reads an organization's HR system, issue tracker, calendar and documents, works out which deadlines and meetings an employee's leave affects and who can cover them, and drafts a coverage plan for a person to approve. The systems are real instances of Frappe HR, Jira and Google Calendar; the organization inside them is synthetic, generated together with its answer key.
- **Evaluation:** In a comparison frozen before the data existed, the agent answered **24 of 30** scenarios correctly in full where a single call shown every document answered **14** (+33 points, 95% confidence interval 20 to 47), at the larger of two corpus sizes. Most of those single calls stopped at the 8,192-token output maximum, which applied to every model call in the comparison (Claude Haiku 4.5 throughout). At the smaller size the single call scored higher, 30 to 24. In the agent's six misses every document it needed had been returned to the run; the failures came after retrieval, in stating the facts.
- **Reliability and control (PostgreSQL, AWS):** The model states facts with their quotes, and deterministic rules draw every conclusion. Every step of a run is an event in one PostgreSQL log, from which crash recovery, the audit trail and the public replays are read. Each run reserves its budget before its first model call, and the answer key sits where the application's AWS role cannot read it.

SteamLens — evaluation-first LLM review-analysis app Jul – Sep 2026

Python · FastAPI · SQLite · multi-provider LLM APIs (DeepSeek, Gemini) · Docker · GitHub Actions · Cloudflare
steamlens.ardabasarici.dev · github.com/arda-basarici/steam-lens · original evaluation report (PDF)

- **Application:** Turns a game's Steam reviews into an evidence-backed report of what players praise and complain about, every finding tied to source reviews or to statistics computed from them. Shipped solo, with CI evaluation gates and approval-gated deploys; a 135,260-review census was labeled for $3.80 at the time.
- **Measured quality:** The deployed LLM labeler, which marks the game aspects each review discusses, was re-certified at **F1 0.801 (95% CI 0.752–0.844)** against a 250-review gold set (one annotator adjudicating over model drafts) after the provider re-pointed the model id to a new model. The original request format lost about a quarter of its labels to parse failures on that model and was replaced; a version check now stops a run at the first response from an unexpected model.
- **Guardrails:** Quotes are verified verbatim against the review text when stored (~2.9% of attempts rejected), and a gate blocks any unverifiable numeral or quote before a report is published. Attribution errors remain, and their measured rate is disclosed inside the product.

Platform — infrastructure and operations under both applications Aug 2026 – Present

Terraform · Ansible · AWS (EC2, SSM, IAM OIDC, CloudWatch) · Cloudflare · Caddy · Docker · SOPS/age
github.com/arda-basarici/platform

- **Infrastructure as code:** The Cloudflare DNS records and security rules and every AWS resource under the agent are in Terraform; the VPS build is an Ansible play with an acceptance script, proven on a blank host.
- **Recovery and alerting:** A rebuild-and-restore drill took a blank throwaway host to SteamLens serving its restored production backup in about an hour. A restore check runs monthly. Forcing the AWS instance's alarm found and fixed a silent SNS policy refusal.

Earlier projects — ML and data studies, each with a written report Jun – Jul 2026

reports on ardabasarici.dev · code on github.com/arda-basarici

- **Blackjack RL (PyTorch):** A tabular agent (~93% agreement) and a DQN (~91% with a stabilizer stack, still below the tabular agent) audited against provably optimal basic strategy, on a from-scratch simulator validated against the published 0.45% house edge.
- **Learned A* heuristic (scikit-learn):** At the report's 50/50 maze mix, 17.3% fewer node expansions than Manhattan distance at a 0.2% mean optimality gap on held-out mazes.
- **Steam Reviews (pandas, pandera):** A resumable, schema-validated pipeline over 298,553 reviews; each reported finding had to reproduce within individual games.